Nonlinear Dynamics in Computer Graphics: Reaction Diffusion Systems in Procedural Generation

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Abstract

[TODO: Abstract goes here.]

1 Introduction

1.1 Objective of this report

This report aims to study the solutions of Reaction Diffusion Systems and explore their far reaching applications outside of mathematics and chemistry. This study will be used in developing a python program that is capable of producing Turing Patterns.

1.2 Structure of this report

The report has been organized to start with a generalist introduction to Reaction Diffusion Systems followed by a mathematical overview of the topic. In particular, we will investigating the solutions of this class of partial differential equations using stability and bifurcation analysis. Studying the solutions of this project is perhaps the most interesting part of this project as they are partly the reason for choosing this topic. The second part of this report will explore the applications of Reaction Diffusion Systems in a variety of domains such as biology, ecology, physics, etc. The final part of this report will demonstrate original contribution from everything that has been learned in the research and conclude with a summary of the topic and the original contribution.

2 Background

2.1 Procedural textures and Perlin noise

Typically, Procedural Generation is done using a technique called *Perlin* Noise developed by Ken Perlin at the Mathematical Animations Groups Inc. (MAGI) while working on the visual effects for the 1982 Disney Sci-fiction film *Tron* Perlin 1985.

The technique uses *interpolation* to geographic terrain and surfaces such as wood and marble. However, Perlin Noise is random and irregular.

There is no way for the artist to precisely control the output to produce organized patterns.

Is there a technique that is capable of producing organized patterns?

2.2 Reaction Diffusion systems

A reaction diffusion system is a *Nonlinear Dynamical system* that models how two or more chemicals *react* and *diffuse* in space and time. A reaction of two chemicals results in new

substances being formed while diffusion occurs when a chemical spreads from a region with lower concentration to higher concentration. [Citation Needed]. General Form of a Reaction Diffusion System:

$$\frac{\partial \mathbf{q}}{\partial t} = D\nabla \mathbf{q} + R(\mathbf{q}), \tag{2.1}$$

$\mathbf{q}(\mathbf{x}, t)$ is the vector of concentrations for each chemical at position $\mathbf{x}$ and time t, $R(\mathbf{q})$ is the reaction component of the sytem , and D is a matrix representing coefficient of diffusions. The Laplacian ∇ compares the concentration of chemicals around a neighbourhood in space and pushes the molecules of a chemical to lower concentration.

Reaction Diffusion Systems can be of multiple components. Components are defined by how many spatial dimensions are used. For instance, a single component RD systems only models the phenonmenon against one space dimension and time. Two component models are needed for producing pattenrns as shown by turing [citaion needed]. There are also many kinds of reaction diffusion system models. In this report, we will be focusing on a simple RD System named the Gray-Scott Model. (Section 2.4).

2.3 Turing patterns

The idea of this subsection is to discuss a research paper from Alan Turing when he was discovering pattern formation in an Embryo Turing 1952. The paper leads to a result called *Turing Instability* which proposes that adding diffusion to a Stable Nonlinear Dynamical System could produce interesting spatial patterns called *Turing Patterns* [Need to write in detail].

2.4 The Gray-Scott model

The Gray–Scott model was introduced as a model of an autocatalytic chemical reaction [citaiton needed here]. in a continuously fed, well-mixed tank (Gray and Scott 1984). The chemical rule proposed in the gray scott model is a simplification and isn't accurate in real life.

$$U + 2V \to 3V, \qquad V \to P, \tag{2.2}$$

The

$$\begin{aligned} \frac{\partial U}{\partial t} &= D_u \nabla^2 U - UV^2 + F(1-U), \\ \frac{\partial V}{\partial t} &= D_v \nabla^2 V + UV^2 - (F+k)V, \end{aligned} \tag{2.3}$$

- U and V are the chemicals in this reaction and their values correspond to their concentration.
- D_u and D_v represent the diffusion coefficient of each element
- F is the feed rate of U for V.
- k defines the rate of decay for V.

Pearson 1993 did a numerical simulation of the Gray Scott Model and found spatio-irregular patterns [Need to expand].

3 Original Contribution

Stability Analysis

The Gray-Scott system is a nonlinear dynamical system whose equations cannot be solved analytically. Therefore, we consider a method known as stability analysis which will predict the

behaviour of the system and precisely identify the set of values for the parameters F and k that lead to the turing patterns we hinted in the previous section. The paramteter values are found using bifurcation analysis

Steps I'll be doing for stability analysis:

- Firstly, I will drop the diffusion terms to show that the system is uniformly stable.
- But before we do that, I solve the system for it's equillibrium points.
- Then classify the system by substituting equillibrium into the jacobian.
- Is it stable or unstable?
- The second part of this analysis will explain what happens when we add back the diffusion term.
- Does the system become unstable? If it does we do a bifurcation analysis to find which parameters F and k will result in turning patterns.

Numerical Simulation of the Gray Scott Equation

This subsection will document my attempt at a numerical simulation of the Gray Model. The numerical simulation is done using a finite difference approximation using 5 point stencil.The code for the simulation is in the appendix.
I will be using a Finite Difference Approximation scheme based on a 5 point stencil to approximate the value of the partial derivatives at each cell in a 20 by 20 grid size.

4 Conclusions

References

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